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Modelling multiple objectives of land use for sustainable development

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Abstract

Sustainability is a human-centred concept that comprises multiple aspects and objectives of different interest groups. Sustainable development is not readily measurable, except as a compromise between different parts of society, of which some may try to represent future generations of mankind. To determine a sustainable development path in the relationship between agriculture and its natural environment, a profound knowledge of this complex system and its behaviour under different socio-economic conditions is necessary. We present a modelling system which consists of a set of hierarchically linked modules. These modules describe production activities in a way that allows an economic and ecological analysis of these techniques. The heart of the modelling system is a multiple goal linear programming model, which is generated by data base modules. Simulation of single farm models as well as regional models based on simultaneously optimised farm types is possible. The modelling system appears to be a highly flexible tool with respect to the number and type of farms, sites and production techniques. Environmental objectives can easily be included and different levels of goal achievement can be simulated. It is well suited for single farm analysis as well as for regional models, for static as well as dynamic approaches. It allows rapid adoption of the model and rapid calculation of scenarios. Therefore, it is suited for use in interactive environments with users which are interested in repeated runs with little changes in the goal function, prices, subsidies or technical coefficients. The results can be used for policy decisions as well as the strategic planning of individual farmers. Applications of the modelling system will be presented in following papers. In this paper the kind of information the model can generate and the circumstances of their usage are shown. © 1999 Published by Elsevier Science Ltd. All rights reserved.

Keywords: Interdisciplinary research; Sustainable development; Agroecosystem management; Multiple goal linear programming

1. Introduction

1.1. Research issue

Since the last reform of the agricultural policy of the European Union (EU) the objectives of agricultural land use have been extended beyond providing income by producing food and fibre, because of increasing concern for the environment. That has been expressed in the concept of sustainability, comprising a mixture of multiple long and short term goals, such as the objectives of the farmers and social, economic, and ecological objectives of society at the national and

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international level (Neher, 1992; Christen, 1996). Many efforts have been made to integrate nature conservation and environmental protection into agriculture. These, mostly regional, activities have resulted in regulations that are often costly because they are in contradiction to EU-wide regulations (Bauer, 1994; Weinschenck, 1994). A number of studies have been carried out to examine the possibilities to realise single environmental goals, but most of these have not resulted in a sustainability oriented agricultural policy (Ganzert, 1995). What is lacking is a concise operational concept of sustainability, possibly because:

1. the concept of sustainability comprises multiple, often conflicting goals that are not clearly defined in terms of measurable parameters;
2. there is no consensus on the parameters that could serve for evaluation of the degree of sustainability of land use, and how the necessary trade-off between these parameters is to be valued; and
3. the complexities of interactions between social, economic, and ecological systems make it difficult to predict how land use should be changed to achieve a desired level of sustainability.

1.2. A society decision

As land use is a dynamic process governed by social and economic conditions, it is important to understand the relationship between socio-economic conditions and land use, and then adjust the socio-economic conditions in a way that allows land users to make the right strategic, tactical and operational decisions at every point in time (Ganzert, 1995). An appropriate economic frame may be seen as a necessary but not sufficient precondition for sustainable development. To identify such an economic frame, different decision makers at different governmental and non-governmental levels have to make a common effort to formulate solutions, e.g. agricultural policies or market mechanisms. We agree with Röling (1994) that this requires communication and is only attainable if some kind of institutionalised driving force for sustainable development can be established. An organising and controlling institution should therefore co-ordinate the following procedures:

1. consent- or compromise-finding procedures about criteria (e.g. indicators, threshold values) of sustainability;
2. consent or compromise on procedures and instruments to achieve sustainability;
3. applied scientific research to identify criteria for sustainability that generate relevant information for decision procedures; and
4. applied scientific research on the interactions between economic and ecological aspects of the system; the results should show how different economic frames affect the decision making of farmers which in turn affects ecological values.

Within this process the task of social scientists, economists, agronomists, and ecologists is to provide appropriate methods and to identify the possible range of choices and consequences. We will show in this paper how our modelling system can be used in the process of the development of sustainable land use options. Following Grant and Thompson (1997) who stated "...that the most reliable way to modify human behaviour for effective natural resource management is to bring key decision-makers through a process of discovery...", we want to contribute to that process by providing appropriate information about land use options (Yiridoe and Weersink, 1997). According to Brandes and Woermann (1982) it is possible to model these complex systems to examine the decision behaviour of the farmers and its most important effects. The model should aggregate the results in such a way that possibilities and limitations, relationships, and interdependencies become explicit. Especially important are conflicting goals to show the trade-offs among the multiple goals that contribute to sustainable agriculture (Romero and Rehman, 1989; van Kooten, 1993). This paper deals with our modelling system, its assumptions, applicability and its shortcomings.

1.3. The modelling approach

There is a need for a modelling tool to analyse agricultural sustainability as a combination of economic and ecological objectives. This model should be able to:

1. simulate effects of political and economic conditions on decisions about agricultural land use at farm level;
2. screen current and new production technologies in a standardised form and show their effects on defined indicators of sustainability; and
3. allow economic and ecological evaluation of production techniques at regional scale, including trade-offs among ecological and economic goals with respect to one farm or to a group of several farm types (regional approach).

Development of this tool still continues and it will become a family of different modules, which together we call Multi-Objective Decision support tool for Agroecosystem Management (MODAM). The model presented in this paper simulates agricultural decision making as economically motivated, and some consequences for the environment.

1.4. Research areas

As this approach tries to unify different aspects and scales we developed the modelling system concurrently for three research regions and projects at three different levels of aggregation:

1. Parts of the Man-And-Biosphere reserve, 'Schorfheide-Chorin' (Erdmann and Nauber, 1995), with about 40 farms and 20 km^2 of the research area serving in the first place as verification of the ecological modules of the model system. Several farms are aggregated to regional model farms to evaluate the effects of different protection strategies and the methodology of developing region-farms.
2. In the National Park "Unteres Odertal", about 32 farms with a total area of 200 km^2 are modelled. The farm models developed are highly detailed and used for verification of the economic parts of the model system. Here, a regional model is used that allows individual farms to interchange their fields and determination of the most economic placement of the total reserve of the park.
3. The third region—the 'Bundesland Brandenburg'—which contains the other two regions mentioned, stretches over 52,000 km^2, serving as a basis for the development of the regional model. The agricultural production activities are defined in terms of representative farm models. Different scenarios of agronomic and ecological political decision making will be developed.

2. Multiple objectives of sustainable land use

First we define a suitable reference system for the assessment of sustainable development. This reference system is based on measurable indicators for each of the objectives pursued. The second part contains some theoretical reflections on possible linkages of multiple ecological and economic objectives with the help of models generating trade-off functions.

2.1. A reference system for sustainability

A necessary precondition for any evaluation is the definition of criteria. For evaluation of actual or proposed land use, various criteria or indicators constitute a reference system, reflecting the different objectives of sustainable land use (Bork et al., 1995; Jansen et al., 1995). For each indicator a minimum level has been defined, following the save-minimum-standard approach of Ciriacy-Wantrup (1952). Obviously, a land use system can be called sustainable only if all criteria are satisfied. Since these indicators represent all criteria for sustainable development they should cover a whole range of single goal parameters in one value, and should be easy to model so that their values can be evaluated for different scenarios. To allow evaluation of goal achievement the indicators have to be measurable at low cost (Ciriacy-Wantrup, 1952).

2.1.1. Indicators of sustainability

Since sustainability refers to both the abiotic environment and the biota, a distinction can be made between abiotic and biotic indicators. Abiotic indicators include nitrogen leaching, soil erosion, etc. For some of these indicators, as nitrogen leaching, statutory fixed minimum standards have been developed for some areas. For others, like soil erosion, normative levels can be identified. However, standards for biotic indicators that reflect the quality of the living resource base in agriculturally used areas hardly exist (Bastian and Schreiber, 1994). As it is not easy to identify generally valid indicators with respect to the type and/or number of species or the number of individuals of one species that should be present in a certain landscape, some ecologists argue that it is sufficient to create suitable physical conditions for the required biological diversity (Roweck, 1995). If the need for the presence of certain species has been accepted, scientists can define the minimum size of adequate habitats or the acceptable level of land use intensity for these species (Bishop, 1978).

However, the extent and type of space for nature protection and development is a societal decision that also depends on what can be afforded. So far, no consistent quantitative system has been defined for sustainability that reflects in a coherent way all economic and ecological aspects. In each of the research projects in which the authors are involved, indicators of sustainability have been identified, but up to now no consistent set has been established. Further research is necessary to develop a flexible reference system, that allows easy adoption to different regions and different demands of society. In this paper we will not elaborate on the development of indicators but focus on the methodology of analysis of a set of indicators of economic and ecological sustainability.

2.2. Economic and ecological analyses

Sustainable development aims at preserving private and public goods to secure future survival of mankind with high probability. It is assumed that the natural environment as a future means of production has to be preserved as healthy as possible. Sustainability is by definition a compromise aiming at concurrent realisation of various objectives. Here, we show how economic and ecological analyses of land use systems can be linked. In our opinion this requires integration of ecological objectives into economic activities (Ciriacy-Wantrup, 1952; Pearce and Moran, 1994). We first discuss the basic economic considerations of finding an approximate equilibrium of demand and supply for natural public goods. Subsequently, we show how this process can be supported by means of trade-off functions.

2.2.1. Economic analyses of market equilibrium for private goods

Assuming a perfect market for private goods, the tangential intersection of the transformation curve and the indifference curve represents the market equilibrium (Fig. 1). In theory, the market determines the production level through the equilibrium between demand and supply. This equilibrium reflects the optimum combination of different goods produced and the optimum allocation of scarce resources for society (Henrichsmeyer et al., 1991).

2.2.2. Non-market products

The goods dealt within environmental and nature protection are often public goods which in the short term are available without any restrictions and any costs for the private users. Nevertheless, use of these goods causes costs and benefits for human society (Tisdell, 1993), and the social benefit–cost ratio is different from the private one (Gowdy, 1993). This situation results in positive or negative external effects known as a type of market failure (Baumol and Oates, 1988; Bromley, 1991). Market failure always leads to imperfect resource allocations and loss of benefits to society as a whole. Economic activities have to integrate costs and benefits that originate in natural or environmental non-marketable goods (Pearce and Turner, 1990). Internalisation of external effects can only cover those aspects of sustainability that are related to external costs.

The consequences of incomplete information in this process have to be taken into account (Weimann, 1991). For ecological goods the demand side often only has vague ideas about

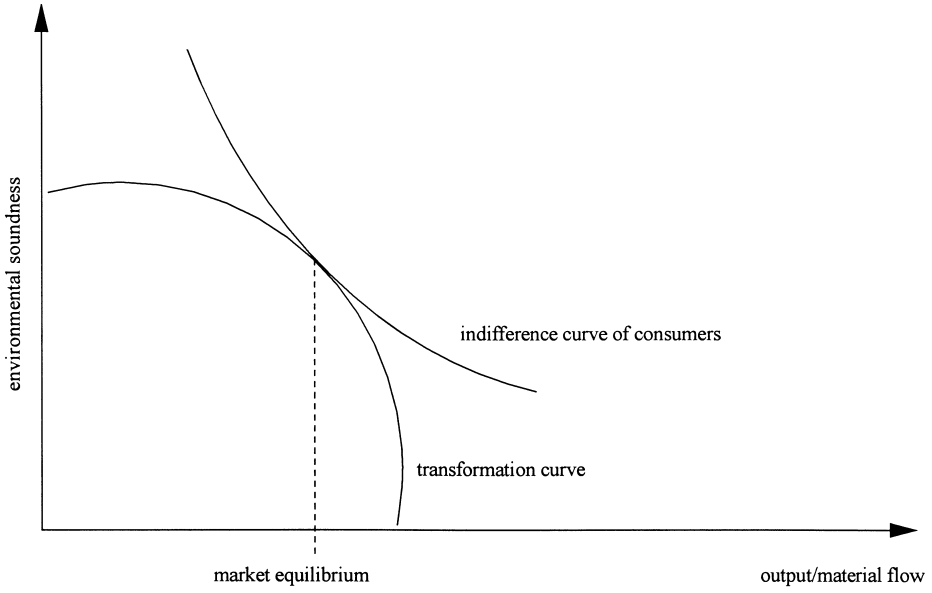

Fig. 1. Market equilibrium, found at the tangential intersection of the transformation curve and the indifference curve.

the net benefits achieved from pursuing ecological objectives. Our modelling system does supply information on ecological and economic costs, and benefits of adapted agricultural practices under different scenarios with respect to socio-economic conditions.

2.2.3. A market for non-market products

In the process of sustainable development, government as well as private activity should first try to create a situation of nearly perfect information, not only about the private, but also about the social transformation curves (Tisdell, 1993). This means that the private and social costs and benefits of the use of these common, non-market goods are known. This information would support a bargaining process among interest groups (Röling, 1994). The bargaining process can be regarded as the market for non-market products, which results in decisions about the extent to which the use of common goods is tolerable for society and whether and what instruments should be used to achieve that extent (Buttgereit, 1991).

2.2.4. Trade-off and indifference curve

A well-established method to provide adequate information for multiple objective problems is the multiple goal linear programming analysis (van Keulen and Veeneklaas, 1992). The trade-off between two conflicting goals shows how much of one objective has to be sacrificed to more fully realise the other objective. Modifications in technical coefficients due to research or in economic coefficients due to changes in markets, etc., can be taken into account by recalculating the trade-offs among objectives. According to Assfalg and Werner (1992) the trade-off function between two objectives, generated by maximising the use of limited resources, can be assumed to be identical to the production-possibility or transformation function and to the supply function. Given consumer preferences for the two objectives, the indifference curve can be added. In this way, the market equilibrium for non-marketable products can be found analogous to that for marketable products (Fig. 1).

As different interest groups or stakeholders have different preferences, they also have different indifference curves (Fig. 2). The above mentioned bargaining process should result in a compromise represented by the intersect of the trade-off or supply function with the indifference function of society (Fig. 3), which can be considered as the market equilibrium for non-marketable products.

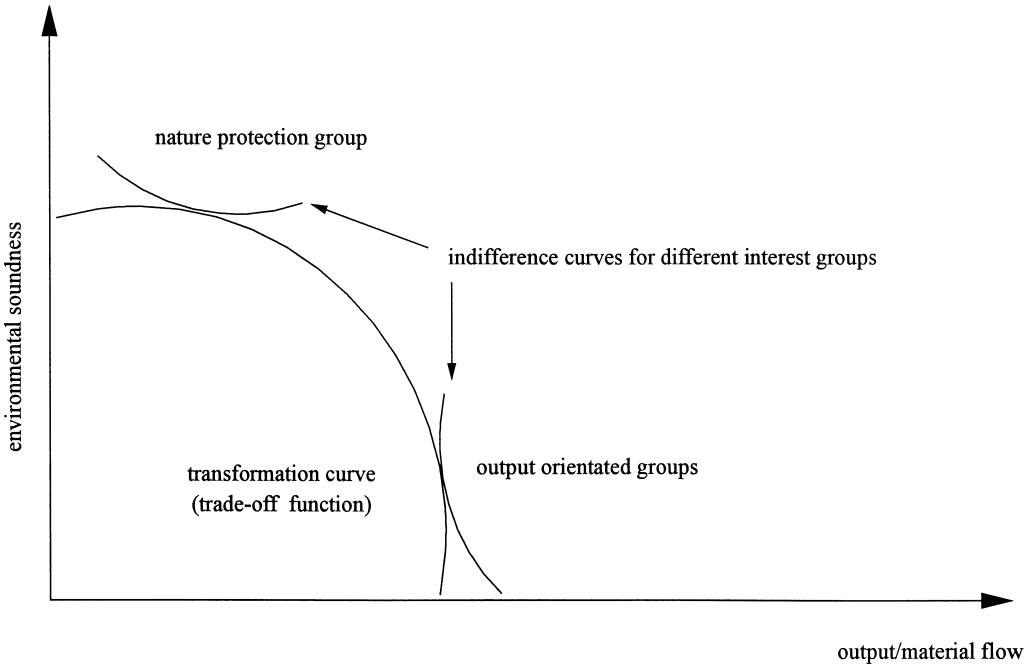

Fig. 2. Different interest groups presented by different indifference curves.

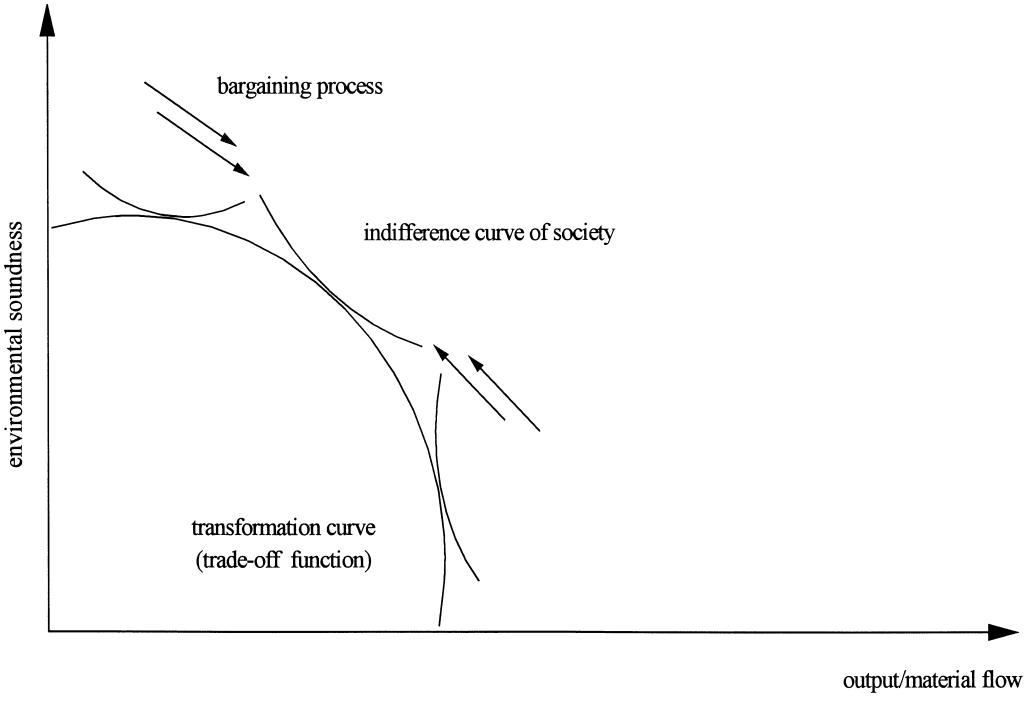

Fig. 3. Compromise-finding procedures should result in a unique indifference curve of society.

As the regional stakeholders are presented by a limited number of organisations, they can be interviewed with respect to their demand for ecological functions. Confronted with the trade-off function (production function) the intersection of their demand functions (indifference curves) with the trade-off function can be determined.

2.2.5. *Bargaining on the basis of trade-off functions*

On the basis of an explicit trade-off function it is easy to imagine how a bargaining process can take place among the different stakeholders involved in the ecological functions represented by the private and public interest groups. If a consensus is found about a common indifference curve, its intersection with the trade-off function can be assumed to represent the theoretical equilibrium between supply of and demand for ecological functions. This process becomes more complicated and leaves less space for bargaining if multiple objectives are involved. The methodological approach however still remains the same.

2.2.6. *Multiple objectives*

As sustainability by definition deals with multiple objectives, the intersection of the indifference curve with the trade-off function of each pair of objectives has to be determined. Trade-off functions for alternative land uses can be competitive, complementary, supplementary, and antagonistic (van Kooten, 1993). Mathematically, it is no problem to develop multidimensional trade-off functions. In contrast, for visual presentations only pairs of goal values are practicable. The number of pairs of objectives is equal to $0.5n\times(n-1)$, with n the total number of objectives. If the number of objectives is high, as for example in Bork et al. (1995), it may be better to present the solutions by means of a limited number of scenarios, generated by combining a number of objectives in each scenario.

2.2.7. *Sensitivity analysis*

As trade-off lines are the result of a mathematically exact, but very simplified, model of a complex reality which relies heavily on expert knowledge, errors of different kind may occur during input data collection. Hence, it is necessary to investigate the effects of such errors on the trade-off functions by extensive sensitivity analysis of the technical and economic coefficients (Paris, 1991). The results indicate which coefficients most strongly affect the behaviour of the whole system and should therefore be estimated with special care. It will also lead to recommendations for further research.

2.2.8. *Scenarios based on different goal achievements*

Each point on the trade-off function refers to the same frame conditions with respect to the goal function. At the starting point, the demands on all goals are set to the save-minimum-standards according to the reference system. Then the restriction on one of the goals is tightened step by step. These different runs can be considered as one set of scenarios. Together with other sets of scenarios that illustrate model behaviour if other goals are restricted gradually, a scenario family is formed, calculated under the same socio-economic and technical conditions. The results can be shown in pair-wise trade-off functions that can be considered as transformation functions given certain technical and economic coefficients. The procedure has to be repeated for modified technical or economic coefficients (Bishop, 1980).

2.2.9. *Scenarios based on different conditions*

An alternative way to use scenarios is to identify different sets of conditions and to evaluate the values of all objectives for each set of conditions. Thus, scenarios of land use can be used to show the consequences of different strategies with respect to political instruments and technologies for environmental protection and nature conservation. But, as the model does not simulate the dynamic development of the corresponding markets, results of these investigations of different agricultural policy instruments should be interpreted with care (Tisdell, 1993).

2.2.10. *Interactive evaluation of objectives*

It is possible to establish an interactive procedure with the help of the modelling system, which allows the user to formulate his own

scenario, characterised by different indicator values and reformulating his scenario by 'playing' with the model (Romero and Rehman, 1989; Pitel, 1990). This will give the user a better understanding of the behaviour of the real system and, consequently, also some understanding for the positions of other interest groups in the bargaining process.

2.2.11. Conclusions

In summary, we can conclude that an interactive modelling system, generating trade-off functions between ecological and economic objectives is helpful from five points of view:

1. interactive experimentation with the model increases knowledge and understanding of the behaviour of the real system, but also the understanding of conflicting positions of interest groups in land use;
2. it allows analysis of the maximum goal achievement possible under given conditions and, hence, defines the boundaries of the multidimensional solution space;
3. trade-off functions will show areas of complementarity between goals and areas where a small decrease in achievement of one goal leads to much larger realisation for another goal; this will allow simplification of the bargaining procedures by showing the consequences of changing preferences: movement of indifference curve and new transaction;
4. sensitivity analysis of the model will show where further research is necessary; and
5. scenarios of different conditions will help political decision makers to identify the most efficient instruments to realise the desired goal achievement in practise.

The model, therefore, is a useful tool in an interactive procedure for the definition of sustainability criteria.

3. Description of the model

Here, we describe the theoretical assumptions and the practical implementation of our decision support model. We show hierarchical linkage between the economic and ecological parts of the model (Jarosch, 1990). Where maps in geographical information systems (GIS) are available, the results of the model in the form of crop rotations and their technical, economic, and ecological coefficients can be transferred to the GIS for graphical presentations.

3.1. Boundaries and assumptions

It is important to distinguish between the image of reality in our minds and the mathematical model we construct as an abstract of the perceived reality. On the one hand there are all the elements of reality. However, only a limited number of these elements is selected to describe the relevant aspects of reality for a given purpose within our mathematical model (Bossel, 1992). We assume that the selected elements comprise all essentials to describe the behaviour of the system for the given purpose.

3.1.1. Assumptions

It is assumed that the managers of agricultural enterprises behave fully rationally and that their decisions on the selected production techniques are based on economically optimal solutions of the allocation of their resources under given economic conditions (Henrichsmeyer et al., 1991). The optimal organisation of a farm is, therefore, determined by the given natural, economic, and legal conditions. At the same time, these basic conditions determine the degree of sustainability of a certain land use. Hence, a deterministic relation between economic conditions and sustainability is assumed. The decision on the extent to which sustainability criteria are taken into account is for this reason exclusively determined by the demand side. We are aware that this approach does not fully reflect the real situation, and that environmental knowledge and awareness of farmers can have a great impact on the sustainability of agricultural practices (Weimann, 1991; Leeuwis, 1993). It is further assumed that it is possible to characterise the ecological soundness of agricultural practices on the basis of some indicators. The selection of indicators is, therefore, decisive

for the final selection of sustainable land use systems (Plachter, 1992).

3.1.2. What the model does and does not

We try to simulate the influence of prices and policy regulations on the decision behaviour of farmers and the effect of the resulting agricultural practices on the indicators of sustainability. Prices and policy regulations are the driving forces of the model. The model is static and refers to a partial equilibrium situation which means that neither the year-to-year variability of the weather nor the influence of farmers' behaviour on other farmers or the market are taken into account. The model simulates decision making only on the basis of purely economic reasoning. Therefore, the different skills or the awareness of farmers with respect to environmental risks are not considered. The effects of practical measures on the environment are estimated on the basis of some selected indicators for sustainability described by knowledge-based models. The essential aspects of these indicators are how they are chosen and how their threshold values are defined which should be done in consultation with the relevant interest groups.

3.2. The structure of the model

Theoretical reasoning leads to the conclusion that the best model structure should completely integrate ecological and economic modules (Krayl, 1993). However, the complexity of the selected system leads to a hierarchical structure of economic and ecological modules that can also be applied to derive optimal solutions, if all possibilities are taken into account (Wossink et al., 1992). The modular concept offers further advantages like separate validation and improvement of modules. Our model is based on a multiple goal linear programming approach. It consists of five levels of hierarchically linked modules. The first level of modules generates the technical coefficients. The

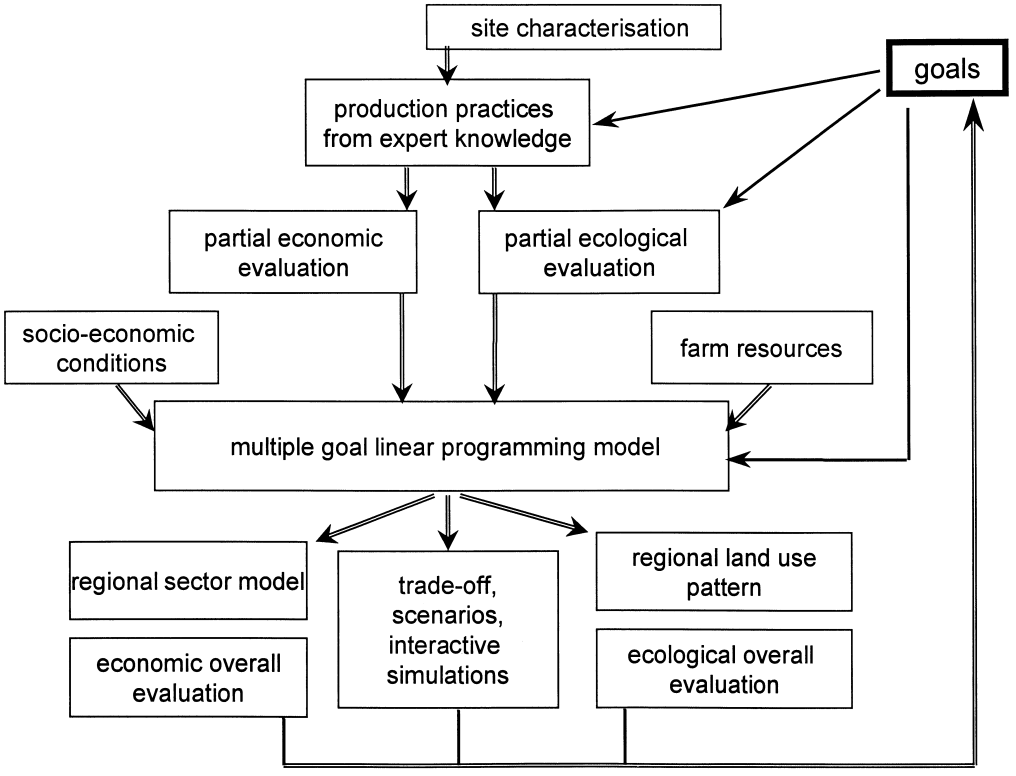

Fig. 4. Iterative definition of goals in the modules of MODAM.

second level calculates the economic coefficients of site-specific production techniques. The third evaluates the ecological effects of these production techniques, and the fourth generates the linear programming model. The fifth level of modules starts the subprogram which solves the equation system, analyses the results and prepares the transfer of data to the Geographical Information System (Fig. 4). Fig. 4 also indicates the interaction between model development and the definition of ecological objectives with respect to the modules: (1) the definition of goal adapted production practices, (2) the formulation of appropriate evaluation tools and (3) the programming of restrictions or goal functions in the linear programming module.

3.2.1. The data bases

The modules were programmed on the basis of a relational data base system which feeds a linear program-solver. It is divided in six different data bases to secure retention of relevant data of individual farms. The modular approach allows different researchers to work on different parts of the model at the same time. Table 1 lists the different data bases developed by different authors (see Acknowledgements section).

The data base CROP generates site-specific production systems based on expert knowledge. The data base ECOL uses the generated data sets from CROP to evaluate some site-specific ecological effects. LIVE contains different sets of livestock production practises, derived from standard data tables. ECON uses the data from CROP, LIVE and from FARM for a partial economic evaluation. The LPGE data base uses information from all other data bases to build a linear programming matrix which is solved by an external solver. The results are returned to LPGE for further ecological analysis and also to prepare the GIS monothematic maps, showing the distribution of single parameters in space for a given scenario.

Table 1
Data bases containing the modules of MODAM

Acronym	Data base
CROP	Crop production practices
LIVE	Livestock production practices
FARM	Individual/model farm data
ECOL	Partial ecological evaluation of cropping techniques
ECON	Standard data[a] and partial economic evaluation of production practices
LPGE	Generator for linear programming models

[a] Standard data are derived from KTBL tables (KTBL, 1997).

3.2.2. Modelling production practices

Modelling production practices has to simulate adequately the operational and tactical decisions with respect to growth of a specific crop at a specific site to allow conclusions on its environmental effects. In the context of our research project it seems redundant to model these decisions at the level of detailed seasonal climatic conditions. Since the project approach is to develop transferable models for the design of new land use strategies, the required data would not always be available. Our approach, therefore, makes extensive use of expert knowledge to generate a set of cropping activities representing long term averages and can be adapted to site-specific techniques with limited data requirements. The resulting standard site-specific cropping techniques are sufficiently accurate for evaluation of environmental goals and not too detailed to make modelling applications for larger areas not feasible. It is possible to design and evaluate prototypes of cropping techniques adapted for more complete realisation of environmental goals.

3.2.3. Crop production techniques

Since this information is implemented in a relational data base, it is possible to link names or numbers of machinery and implements to tables in which working times and costs are specified for each machine or implement. Site-specific amounts of seeds, fertilisers and products are calculated. The expected yield is calculated through equations derived from expert knowledge as a function of site characteristics (Roth, 1995). The fertiliser requirements are estimated as a function of the yield, using the contents of the corresponding nutrient in the harvested components. A large set of possible crop production techniques is defined according to the site-specific potentials. In the farm model, rotations are generated comprising

these techniques, and that meet phytosanitary and technical restrictions.

3.2.4. Ecological evaluation of crop production techniques

The standard crop production techniques have to be evaluated with respect to each indicator of sustainability. Up to now, six evaluation tools have been developed which can be linked to the modelling system. Currently identified indicators are: (1) the potential for nitrogen leaching; (2) potential wind; (3) water erosion; (4) protection of wild flora; (5) a disturbance index for amphibians; and (6) an index for partridges. These evaluation tools are described in detail by Meyer-Aurich et al. (1998). Because of the modular structure of the data base it is also easy to link other evaluation tools.

3.2.5. Multiple goal linear programming farm model

The multiple goal linear programming farm model resembles a common linear programming model with the well-known matrix structure solved by the Simplex method. To handle multiple goals this technique is used with different goal functions. Various series of runs are executed, each time with another objective defined as goal function and the other objectives formulated as restrictions. In the first run, the restrictions are all held at a minimum level defined as sustainable. This solution yields the maximum attainable value for each objective. By repeating these runs with gradually tightened restrictions, the data for the already-discussed trade-off functions are generated (de Wit et al., 1988).

3.2.6. Demand for ecological goals

The production of food and raw materials takes place in EU-wide organised markets and is (co-)determined by EU regulations. Ecological objectives are often regulated at national, regional or local scale. The demand for 'goods' as natural habitats can be formulated by new regulations or by appropriate subsidies restricted to the appropriate spatial scale. In the model, all regulations and prices are exogenous. Therefore, exhaustive sensitivity analyses have to be executed. It is envisaged to optimise these exogenous conditions in an interactive process with decision makers at regional level. The farm model will show where comparative advantages of sites can be used within the farm. The regional model will show the same type of interactions between farms with respect to regional ecological goals.

3.3. The spatial dimension of the model

3.3.1. Regulation area and landscape

Different socio-economic, as well as ecological, aspects of the model are strongly related to defined areas. Farms mostly consist of a number of fields characterised by specific hydrological, soil, and weather conditions. With respect to agricultural production, the fields are the lowest spatial scale within the model. With respect to the ecological evaluation of agriculture, other spatial elements as biotopes and borders to adjacent fields can be added. Fig. 5 shows the spatial dimensions corresponding to the elements of the modelled system. The model will be used for both individual farms and for 'typical' farms in the context of regional modelling. Hence, the relationships in Fig. 5 may refer to individual as well as to types or classes of sites. Subsidies and regulations are designed within the context of definitive spatial boundaries (Scheele et al., 1992). This generalisation refers to subsidies and regulations of the EU, but also at national scale, regulations by the "Bundesländer" and other locally specific regulations, i.e. the possibilities for subsidies can differ from one farm to another. Farms are characterised by their site-specific access to markets, and their access to the production factors land, labour and capital, which are inter alia depending on the location of the farm (Reisch and Zeddies, 1992). Moreover, biological indicators of sustainable land use may differ from one landscape to another; e.g they vary with the potential for rare plants and animals in different types of landscapes. Therefore, land use systems have to be evaluated in relation to the corresponding landscape (Buys, 1995).

3.3.2. Farm typology

At the scale of a "Bundesland", modelling individual farms is not feasible. However, individual farm models should be used for validation of the

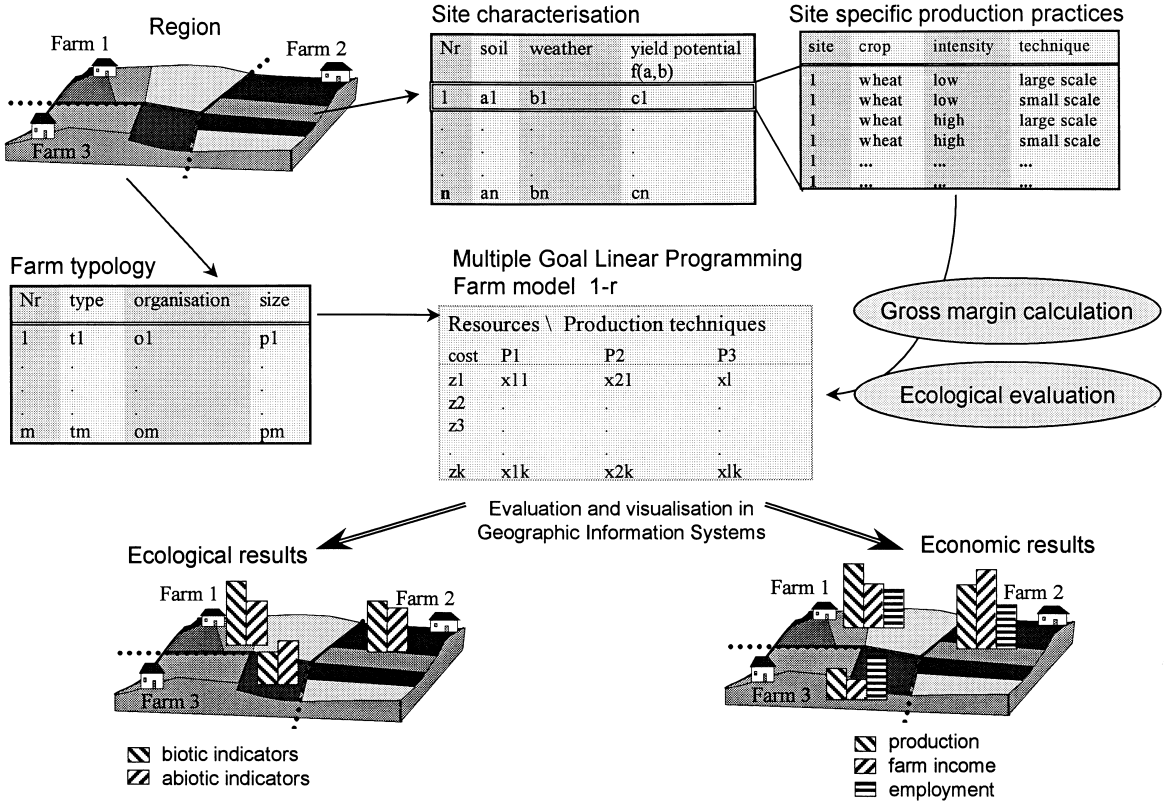

Fig. 5. From site and farm typology to a regional evaluation of land use scenarios.

modelling approach. A typology is necessary to identify classes of farms that can be simulated with the same farm model. The typical parameters of this typology are legal form of the farming enterprise, production orientation and size class. These farm types are differentiated on the basis of the ecological characteristics and the production potential of their sites, which are based on a site classification, necessary to handle the data needed for large scale models.

3.3.3. Site classification

The classification of soil types forms the basis for all other typologies. For characterisation of sites in former Eastern Germany, two types of soil maps at two levels of aggregation are available. First the "Reichsbodenschätzung" (National Soil Quality Estimation) that was conducted in the thirties for reasons of tax estimation and resulted in a very detailed picture of the soils in all of Germany before Word War II. The data for each polygon of a few hectares were transformed in a soil fertility index, indicating the production potential of the soils. After World War II in Eastern Germany the "Reichsbodenschätzung" was transformed into a more aggregated map (scale 1:25 000) with polygons of some 10–100 ha classified according to soil quality, hydrology and relief. This so-called Mittelmaßstäbige Kartierung (MMK = medium-scale cartography) is available in digital form and constitutes the second source of soil data (Schmidt, 1978; Gunia, 1982).

3.3.4. Site classification for estimating environmental effects

A first level classification of the environmental effects of agricultural production has to be based on soil classification, which indicates the type of 'vertical' problems to be expected; e.g. nitrate leaching potential. The second level of classifica-

tion should take into account lateral relationships caused by the production on fields. This includes possible impacts of all kinds of emissions from fields to neighbouring structures such as open water, hedges, or woods, as well as disturbances of special habitats; e.g. runoff contaminated with pesticides and fertilisers should be avoided as far as possible. A third type of lateral relationships is given by the landscape in which the site is embedded. Within the Schorfheide-Chorin project, for example, amphibians were chosen as a group of animals that should be protected. Hence, fields in the neighbourhood of certain types of surface water have to be cultivated using special cropping techniques developed for this objective (Meyer-Aurich et al., 1998).

4. Application of the model

The model will be used in different research programs executed at the Centre for Agricultural Landscape and Land Use Research (ZALF), Müncheberg in Germany. Currently, the model system is developed in the context of three research projects:

1. in the biosphere reserve 'Schorfheide-Chorin' within the framework of the combined BMBF–DBU[1] project for the design of land use systems especially aiming at integration of nature protection objectives;
2. in the National Park 'Unteres Odertal' for the analysis of the economic effects of the return of agricultural land to reserve area; and
3. in the 'Bundesland Brandenburg' to investigate the impacts of large scale transformation from conventional to organic farming.

4.1. Research project 'Schorfheide-Chorin'

The project in the biosphere reserve 'Schorfheide-Chorin' aims at developing an agrarian landscape management by mixing agricultural and nature protection objectives. This includes some segregating measures, like letting fields lie fallow, but the main focus is on integrative measures, like the development of adapted cropping practices that take into account the special requirements of naturally occurring species. Both strategies will be evaluated with the help of MODAM. Trade-offs will be calculated between the different nature protection and farmers' goals for model farms. The results should support decision making within the reserve authority, which has to establish efficient nature protection strategies with limited resources.

4.2. Research project 'Nationalpark Unteres Odertal'

The national park was established in 1995. Until 1989 this region was one of the most important grassland areas in the former Democratic Republic of Germany (GDR), used at a high level of input intensity. The major part of the park is still in agricultural use. Objectives of the national park, such as reducing intensity and abandonment of production are competing with the agricultural activities. The national park authority manages subsidies from the EU life programme, the riparian zone programme of the Federal Republic of Germany (FRG), subsidies of the Land Brandenburg and funds of Non Governmental Organisations. The demand for ecological 'goods' is thus concentrated in the hands of the national park authority. With the help of MODAM, a model for this region is being generated that simulates each farm on the basis of a specific farm model. Compensatory reactions in and among farms (exchange of fields), as well as different scenarios with regard to the location of total reserves without any agricultural use will be investigated.

4.3. Research on large scale transformation of land use to ecological farming

The objective of this research project is to analyse some ecological and economic effects of large scaled transformation of conventional to ecological

[1] BMBF: Bundesministerium für Bildung und Forschung (State Ministry of Food, Agriculture and Forestry). DBU: Deutsche Bundesstiftung Umwelt (German Federal Foundation for the Environment).

farming. The research area consists of the 'Bundesland Brandenburg'. In total, eight indicators for this evaluation were identified. The socio-economic goal system is represented by production, income and employment in agriculture. Ecological indicators are leaching of nitrogen and phosphate, C/N balance of the soil, use of fossil energy and area and quality of natural habitats. Since the research area consists of one Bundesland the relevant decision makers to interact with are not only at local level, but also at the level of the Bundesland. Furthermore, non-governmental organisations were included in the interactive process of defining adequate indicators and their goal values.

5. Conclusion

The modelling system, MODAM, is suitable for interactive research on multi-objective land use issues. MODAM is an instrument that can serve to mediate in conflicts among competing groups of land users, by generating information about economic and ecological effects of particular decisions. It is suitable for ex post analysis as well as for ex ante planning problems. MODAM allows depiction of economic and ecological spatial problems at variable scales. The modular structure permits linkage of additional economic and ecological modules and facilitates inclusion of new scientific knowledge. The linkages between the various modules are organised by means of a relational data base and automated to a large extent. Hence, a large number of scenarios can be analysed in a short time. Finally, MODAM did serve during its development as interface and focus in interdisciplinary research.

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